THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 3: **The World of Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The mass of the iron nail after being kept in copper sulphate solution for long: 1

(a) becomes zero **(b)** increases **(c)** decreases **(d)** stays same

2. In the situation above, R and S are ___ than hydrogen. 1

(a) unreactive gases **(b)** less reactive **(c)** equal **(d)** more reactive

3. A student passes the gas formed on burning a substance through lime water and it turns milky. The substance contained: 1

(a) carbon **(b)** sulphur **(c)** iron **(d)** sodium

4. In $\mathrm{Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu}$ the solid left on zinc is: 1

(a) zinc oxide **(b)** zinc sulphate **(c)** carbon **(d)** copper

5. An iron nail dipped in a solution X becomes coated with a reddish-brown deposit and the blue colour fades. X is likely: 1

(a) sodium chloride **(b)** sugar **(c)** zinc sulphate **(d)** copper sulphate

6. Which statement is correct about graphite and diamond? 1

(a) both are forms of carbon **(b)** both are metals
(c) both conduct **(d)** both are soft

7. Which solution will not react with iron? 1

(a) $\mathrm{CuSO_4}$ **(b)** $\mathrm{ZnSO_4}$ **(c)** HCl **(d)** $\mathrm{AgNO_3}$

8. Which one is not a use of non-metals? 1

(a) making electric wires **(b)** fuels
(c) fertilisers **(d)** breathing

9. **Assertion (A):** An iron nail dipped in copper sulphate solution gets a reddish-brown coating. 1
Reason (R): Iron is more reactive than copper and displaces it from its solution.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** All metals are solid at room temperature. 1
Reason (R): Mercury is a metal that is liquid at room temperature.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the bulb filament made of tungsten, a metal, and not of sulphur? 2

12. Why does hydrogen not form when copper reacts with dilute acids? 2

13. Why is iron more useful than sodium although sodium is a metal too? 2

14. Why are lightning conductors made of copper? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why can a metal piece placed in acid sometimes show no bubbles? 3

16. Explain why magnesium ribbons are cleaned before burning. 3

17. Explain why copper sulphate solution turns pale green when iron is added. 3

OR

Write the word equation for the burning of magnesium in oxygen. 3

18. Why should iron tanks used to store rainwater be coated with paint or zinc? 3

OR

Which metal pieces can safely be tested with dilute acids and why not with sodium? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a science lab, Sunil burns a magnesium ribbon and collects the white ash. In another spoon he burns a little sulphur powder and passes the gas into water. He tests both the ash dissolved in water and the sulphur solution with litmus papers. 5
(i) Name the white ash formed from magnesium.
(ii) What will the solution of the ash do to red litmus?
(iii) Name the gas formed on burning sulphur.
(iv) What will the sulphur solution do to blue litmus, and what does this show about oxides of non-metals?

20. Explain the importance of non-metals in nature, living things and industry. 5

OR

Explain why the classification of elements into metals and non-metals is useful. 5

END OF QUESTION PAPER — Total: 40 Marks